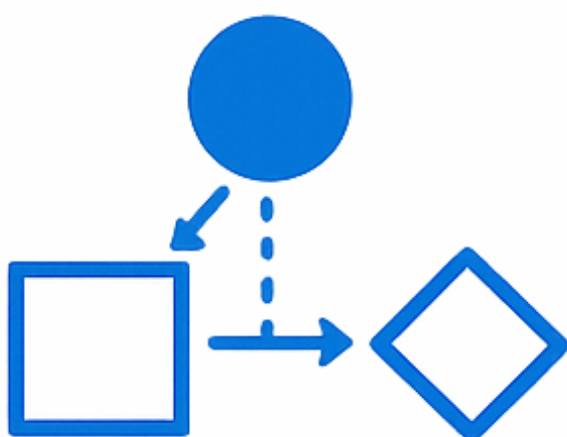




Assignment 2

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Qualitative Methods and Data Quality

Part I: Say-Do Gap Study and Part II: Radiology Data Quality Analysis

Date: March 2, 2026

Course: NDAB20010U Health Data and Interoperability

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AI Declaration

I declare that AI has only been used for grammatical correction.

AI has **not** been used to generate content, reasoning, or structure.

Contents

- 1 Part I: A Real-World Study of the Say-Do Gap in Modelling 4**
 - 1.1 Deliverable 1: Description of the Routine Based on the Interview 4
 - 1.2 Deliverable 2: Event Log Representing the Observed Routine 7
 - 1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice 9
 - 1.4 Deliverable 4: Reflections on the Modelling Process 10

- 2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs 12**
 - 2.1 Deliverable 5: Notebook Submission 12
 - 2.2 Deliverable 6: Part II Reflection Questions 12

- A Supporting Tables for Part I 15**

- B Supporting Tables and Code for Part II 15**
 - B.1 Custom DQ Assessment Method Code Excerpt 16

- References 18**

1 Part I: A Real-World Study of the Say-Do Gap in Modelling

This part analyses one weekday morning routine across *work-as-imagined*, *work-as-done*, and *work-as-recorded*, with explicit links between claims and evidence.

1.1 Deliverable 1: Description of the Routine Based on the Interview

Scope and notation This Part I study uses a single-case self-study design with one participant (the author). The chosen routine is a weekday morning routine.

The interview was conducted on February 18, 2026 as a self-recorded, semi-structured interview with open questions. Following Guide to Health Informatics (2026a), this stage is treated as *manual knowledge acquisition*: it captures *knowledge-as-stated* and establishes a first *work-as-imagined* model of the routine (Guide to Health Informatics 2026; Zajac 2026).

Identifier convention used in this report:

- **Axx** = *activity* IDs from the interview-based model (e.g., A01-A09).
- **Exx** = *event* IDs from the observed event log (e.g., E03, E10).
- **Dxx** = *delta* IDs in the comparison sheet (e.g., D01-D06).
- **Mx** = *method* IDs from the Part II DQ toolkit (e.g., M2, M10).

The model notation *DCR* denotes *Dynamic Condition Response* graph modelling.

The routine boundary is defined as start when the alarm sounds and end when the apartment door closes, based directly on transcript lines L21-L22 in Table 1. The baseline sequence (wake-up, hygiene, dressing/packing, breakfast, checks, departure) is grounded in lines L02-L12, and variation points (snooze, breakfast substitution, key search, occasional roommate coordination) are grounded in lines L14-L18. This line-level grounding makes the interview model auditable.

Routine boundary:

- Start: alarm sounds.
- End: participant leaves home (door closes).

Table 2 presents the interview-based activity model, and Figure 1 shows the corresponding *interview-based DCR* graph exported from DCR Graph. The graph was built from activities A01-A09 and used as the baseline representation for comparison against the observed event log in Deliverables 2 and 3.

Assumptions used for modelling:

- The model represents a weekday baseline (not weekend behavior).

- The participant is the only consistent human actor in the routine.
- Expected times are recall-based interview estimates and are treated as approximate.

These assumptions delimit model scope and support transparent comparison between *knowledge-as-stated* and observed evidence in Deliverables 2 and 3.

Interview transcript evidence Table 1 contains the line-numbered transcript excerpt used as evidence for Deliverables 1-4.

Table 1: Line-numbered self-interview transcript (conducted February 18, 2026)

Line	Transcript
L01	Interview question: Please describe your normal weekday morning routine from when you wake up until you leave home.
L02	Participant: My alarm goes off at 06:45, and I usually get up right away after stopping it.
L03	Participant: I first drink a glass of water, then I go to the bathroom.
L04	Participant: I normally take a quick shower, around ten to twelve minutes.
L05	Participant: After that I get dressed and prepare my bag for the day.
L06	Participant: I usually pack my laptop, charger, student ID card, wallet, and keys.
L07	Participant: Then I make coffee and toast.
L08	Participant: I normally sit down, eat breakfast, and check my calendar and transport timing on my phone.
L09	Participant: After breakfast I brush my teeth and do a final mirror check.
L10	Participant: I check the weather so I can choose the right jacket.
L11	Participant: Before leaving, I do a final essentials check: phone, wallet, keys, and card.
L12	Participant: I aim to leave at around 07:30 to catch transport to campus.
L13	Interview question: Which steps vary the most from day to day?
L14	Participant: Sometimes I snooze once if I am tired.
L15	Participant: Breakfast can change; if I am in a rush, I might just eat fruit instead of toast.
L16	Participant: Occasionally I spend a minute looking for keys if I did not put them in the same place.
L17	Interview question: Are other people typically involved in this routine?
L18	Participant: Mostly no, I do it alone, but once in a while I text my roommate if I need a shared item.
L19	Interview question: Which tools or systems are central in the routine?
L20	Participant: My phone alarm, calendar, weather app, the coffee machine, toaster, and my backpack.
L21	Interview question: How do you define start and end of this routine?
L22	Participant: It starts when the alarm sounds and ends when I close the apartment door behind me.

Table 2: Interview-based activity model (work-as-imagined / knowledge-as-stated)

ID	Activity label	Actor	Order	Expected time	Main tools/resources
A01	Stop alarm and orient to time	Participant	1	06:45-06:47	Smartphone alarm
A02	Drink water	Participant	2	06:47-06:50	Glass, tap
A03	Toilet and shower	Participant	3	06:50-07:02	Bathroom, shower
A04	Dress and pack bag	Participant	4	07:02-07:12	Wardrobe, backpack
A05	Prepare breakfast and coffee	Participant	5	07:12-07:20	Coffee machine, toaster
A06	Eat breakfast and review schedule	Participant	6	07:20-07:25	Phone calendar/transport apps
A07	Brush teeth and final grooming	Participant	7	07:25-07:28	Toothbrush, mirror
A08	Check weather and essentials	Participant	8	07:28-07:30	Weather app, keys, wallet, ID
A09	Leave apartment for commute	Participant	9	07:30	Front door, bag

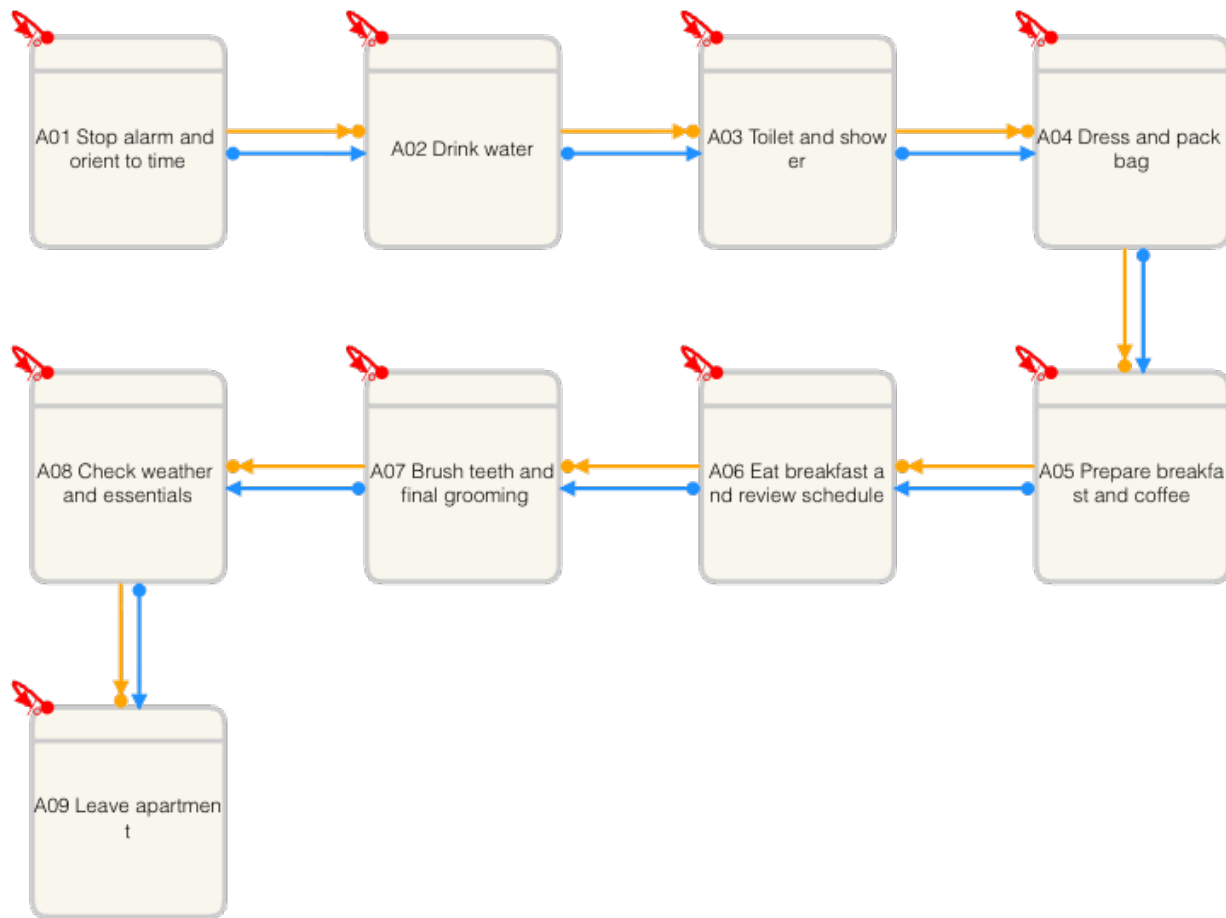


Figure 1: Interview-based DCR graph for the morning routine (A01-A09)

Source: Own DCR Graph export (February 2026), built from transcript evidence in Table 1 (lines L02-L12, L14-L18, L21-L22).

Note: Used as baseline work-as-imagined model for comparison with observed events in Deliverables 2 and 3.

Interview-based activity model

1.2 Deliverable 2: Event Log Representing the Observed Routine

The observation was conducted on February 20, 2026 (06:45-07:30), two days after the interview on February 18, 2026. This separation supports comparison between interview elicitation and situated execution.

As described in Chapter 27 and the course methods slides, *observation* complements *interviews* by capturing situated judgement and practical adaptations that are often missing from recollection-based accounts (Guide to Health Informatics 2026; Zajac 2026). To make data collection explicit and reproducible, a short observation guide was used (Appendix A, Table 5).

Notes were split into *descriptive notes* and *reflexive notes*, and only descriptive notes were transcribed into the event log (Zajac 2026).

The full machine-readable event log is submitted as a separate CSV artifact with timestamps, event labels, and context notes. This file is the *work-as-recorded* layer of the study and the empirical basis for the *observed DCR* comparison in Deliverables 3-4.

The submitted CSV uses the following fields: `case_id`, `event_id`, `timestamp` (ISO datetime), `event_label`, `actor`, `tool`, `context_note`, and `related_activity_id` (A01-A09 or NA).

Table 3 provides the observed event-log excerpt with activity references. Relative to the transcript-based baseline (Table 1, lines L02-L12 and L14-L18), the key comparison anchors are E03 (extra coordination), E09 (order shift), and E10 (workaround).

The observation confirms the same overall routine boundary as Deliverable 1, but adds practical variation and coordination not present in the interview baseline. In particular, one extra coordination event (E03), one order change (E09 relative to A04/A05), and one explicit workaround (E10) are central for the delta sheet in Deliverable 3. These events represent *work-as-done* under actual constraints.

Table 3: Observed event-log excerpt (February 20, 2026)

Event	Timestamp	Event label	Tool	Context note	Ref.
E01	2026-02-20 06:45	Alarm rings and snooze	Smartphone alarm	One snooze before wake-up.	A01
E03	2026-02-20 06:52	Reply to roommate message	Messaging app	Extra micro-coordination event.	NA
E05	2026-02-20 06:57	Toilet and shower	Bathroom	Hygiene block started later than interview expectation.	A03
E07	2026-02-20 07:12	Start coffee and toast	Coffee machine, toaster	Breakfast preparation begins.	A05
E08	2026-02-20 07:16	Switch breakfast to banana	Kitchen	Breakfast adapted due to no bread.	A05
E09	2026-02-20 07:18	Pack bag	Backpack	Packing occurs after breakfast prep starts.	A04
E10	2026-02-20 07:21	Find keys and borrow lock	Keys, messaging app	Workaround to solve missing item.	A08
E14	2026-02-20 07:30	Leave apartment	Front door	End of observed routine boundary.	A09

1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice

The comparison maps interview activities (A01-A09) to observed events in Deliverable 2. A delta was registered when at least one of the following differed: event presence, order, timing, workaround behavior, or interpretation of activity labels. Methodologically, this follows Chapter 27's logic of comparing observed process traces with a pre-existing process description to identify frequent exceptions and model limits (Guide to Health Informatics 2026).

Table 4 summarises the full delta comparison between stated and observed routine. Key deltas are:

- *Missing event*: no distinct seated breakfast-and-schedule-review step in observation.
- *Extra event*: additional roommate coordination event appears only in observed data.
- *Different order*: bag packing occurs after breakfast preparation begins.
- *Timing distortion*: hygiene block starts later than stated baseline.
- *Workaround/coordination step*: missing lock triggers ad hoc borrowing via messaging.
- *Relabeling/ambiguity*: “breakfast” shifts from sit-down meal to banana plus coffee-to-go.

All six required categories are documented in Table 4: missing events, extra events, different order, timing distortions, workarounds/coordination steps, and relabeling/ambiguity.

Overall, the *interview-based DCR* (Deliverable 1) captures *knowledge-as-stated* and *work-as-imagined*, while the observed event sequence (*observed DCR* evidence) captures *knowledge-as-discovered*, *work-as-done*, and *work-as-recorded*. The delta sheet therefore provides traceable evidence of the *say-do gap*.

Table 4: Delta sheet between interview model and observed routine

Delta ID	Category	Interview ref.	Observed ref.	Difference description	Implication
D01	Missing events	A06	No direct event	No distinct seated breakfast-and-schedule-review event occurred.	Interview model overstates routine regularity.
D02	Extra events	None in baseline	E03 (06:52)	Short roommate-message coordination appears in observed routine only.	Reveals practical coordination load.
D03	Different order	A04 before A05	E07 then E09	Bag packing happened after breakfast preparation started.	Ordering constraints are weaker than stated.
D04	Timing distortions	A03 expected 06:50-07:02	E05 start 06:57	Hygiene block shifted later due to wake-up delays.	Time windows in interview are optimistic.
D05	Workarounds/coordination steps	A08 final check	E10 (07:21)	Missing lock triggered ad hoc borrowing via message.	Recovery actions are underrepresented in interview model.
D06	Relabeling/ambiguity	A05/A06	E08, E13	“Breakfast” changed from sit-down meal to banana plus coffee-to-go.	Label definitions need operational clarity.

1.4 Deliverable 4: Reflections on the Modelling Process

1. **Data missed or misrepresented in the interview-based model.** The interview model captured routine intent, but missed situated details that appeared in observation: micro-coordination (E03), recovery behavior (E10), and omission of the planned seated breakfast-and-schedule step. This pattern is expected in *manual knowledge acquisition*:

interviews primarily capture recalled intent, while *observation* captures situated adaptation (Guide to Health Informatics 2026; Zajac 2026). In this case, the interview model therefore overestimated routine linearity.

2. **What counted as an activity in each model.** In Deliverable 1, the *interview-based DCR* uses intention-level activities (A01-A09). In Deliverable 2, the event log (the empirical basis for the *observed DCR*) uses timestamped observable actions. This granularity mismatch means one DCR activity can map to multiple logged events. The same routine can therefore appear stable at intention level but variable at event level.
3. **How the event log constrained what could be modelled.** The event log supports strong analysis of order, timing, and deviations (*work-as-recorded*), but it captures motives and judgement only indirectly. Without contextual notes, causal interpretation of why a deviation occurred remains weak. This is why descriptive/reflexive note separation was retained in the method and documented in Appendix A, Table 5.
4. **Potential and limits of automatic event capture.** Automatic capture is strong for digital traces (alarm, app checks, timestamps), but weak for tacit coordination, local workarounds, and searches for missing items. In Chapter 27 terms, automated traces should be combined with *qualitative discovery* methods when the task includes social and contextual adaptation (Guide to Health Informatics 2026; Zajac 2026).
5. **Variations and exceptions revealed by observation.** Observation revealed concrete *exceptions*: changed order (E07 before E09), omitted planned step (A06 not distinct), and workaround behavior (E10). Applied to this case, Chapter 27 implies that process evidence should not only test conformance but also explain recurring exceptions that may be locally rational (Guide to Health Informatics 2026).
6. **Relevance of qualitative discovery in process modelling.** For this routine, model quality improved by triangulating *knowledge-as-stated* (interview + transcript), *knowledge-as-discovered* (observation), and *work-as-recorded* (event log). The resulting model is more realistic than any single method alone, consistent with Chapter 27's iterative *knowledge acquisition* approach (Guide to Health Informatics 2026).

2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs

This part synthesises notebook evidence (Deliverable 5) and reflection-based interpretation (Deliverable 6) of *data quality*, *conformance*, and *invisible work* in radiology event logs.

2.1 Deliverable 5: Notebook Submission

The completed notebook is submitted as a separate artifact (`assignment.ipynb`), as required by the assignment brief.

The notebook contains the full Part II workflow (Parts 0–7): initial exploration, method selection and execution, custom *DQ dimension* design, *five-facet* synthesis (*Data*, *Source*, *System*, *Task*, *Human*), and *invisible-work* analysis.

Deliverable 6 interprets these outputs as *work-as-recorded* evidence and discusses where log-based findings need qualitative or design-level contextualisation.

2.2 Deliverable 6: Part II Reflection Questions

Notation and framing In Part II, **RQ** denotes *research question* (RQ1–RQ3), and **M** denotes one selected assessment *method* from the toolkit (e.g., M2, M10). The report uses *TAT* as shorthand for *turnaround time*.

- 1. Initial exploration and relevance to the three research questions.** The dataset contains 150 events, 17 cases, 14 columns, and 12 activity types across 2026-01-12 07:45 to 2026-01-13 08:46. Core workflow fields are present, while context-heavy fields are sparse (e.g., `study_uid`, `series_count`, `instance_count`, `icd11_code`). Following the *Five Facets* framework, this means the dataset is stronger for *Task/System* sequence analysis than for rich *Data/Source/Human* interpretation (Mohammed et al. 2025). Applied to this case, the file can support RQ1 (workflow efficiency and *TAT* bottlenecks), RQ2 (*conformance* for critical communication), and RQ3 (*representativity* and workload patterns for modelling), but with explicit limits on contextual inference.
- 2. Method selection justification and trade-offs.** M2, M6, M8 (15 min), M10, and M11 were selected as a *coverage portfolio*: together they cover all three RQs with low redundancy across facets. The completed selection table is documented in Appendix B, Table 6. The core trade-off was breadth across *facets* versus depth in fewer dimensions. RQ1 weighted timing integrity higher (therefore M10), RQ2 weighted communication safety higher (therefore M8), and RQ3 weighted representativity/workload structure higher (therefore M6 and M11).

Non-selected methods were excluded for explicit reasons: M1 (schema validation) added limited incremental value after initial structure checks; M3 (ICD-11 validation) addressed a narrower coding issue than the selected cross-RQ set; M4 overlapped with M11 for actor/role distribution; M5 offered a narrower modality slice than M6; M7 overlapped with M10 plus the custom batch-credibility analysis; and M9 was useful but lower priority than the selected set for cross-RQ coverage.

3. **Results interpretation by method.** Chapter 27 explains that *process mining* uses temporal event sequences to reveal common pathways and frequent exceptions (Guide to Health Informatics 2026). Applied here, Appendix B, Table 7 shows:

- M2: high context missingness (e.g., `study_uid` 134/150), so flow-level findings are feasible but context-sensitive claims are weaker; expected and coherent with synthetic-log design.
- M6: urgency and shift imbalance (15 stat/urgent vs 2 routine; Day 11, Night 6, Evening 0), so *representativity* for broad generalisation is limited; expected and relevant for RQ3 risk.
- M8: 16 pass and 1 fail on critical communication; overall *conformance* is high, but the single fail remains operationally important for RQ2; method behaved as expected.
- M10: 6/17 cases with *batch-logging* signatures and a large raw-vs-clean TAT gap (147.93 vs 198.50 min), making timestamp credibility a central RQ1 risk; this was the most surprising result.
- M11: uneven actor distribution (radiographer 64 events, resident 7, system actor concentration), indicating workload concentration affecting interpretation; expected and useful for RQ1/RQ3 context.

Possible improvements are stricter timestamp validation rules for M10 and role-normalised comparators for M11, to reduce overinterpretation from small samples.

4. **Proposed DQ dimension.** The proposed dimension is *Batch Logging Credibility*: whether timestamp structure reflects real-time documentation or delayed/post-hoc entry. The motivating pattern from exploration code was repeated timestamps and tight clustering in cases such as CXR-013 and CXR-011. In this context, low credibility can bias RQ1 (*TAT* and bottlenecks), mask RQ2 communication delay, and leak artefactual timing patterns into RQ3 modelling. The custom method code is included in Appendix B.1, and the facet mapping is documented in Appendix B, Table 8.

5. **Per-facet data quality synthesis.** The custom method code is documented in Appendix B.1. It scores per-case timestamp risk from duplicate-time frequency, short-window event density, and low unique-timestamp ratio.

- *Data*: missing context fields and timestamp artefacts reduce precision; consequence is weaker causal interpretation in RQ1 and RQ2.
- *Source*: mixed real-time and delayed logging reduces reliability; consequence is uncertainty about what truly happened when.
- *System*: safety-relevant communication events can be missing despite completed workflow; consequence is false reassurance for RQ2.
- *Task*: skewed representativity limits external validity for optimisation/triage; consequence is weak transferability for RQ1 and RQ3.
- *Human*: tacit coordination is weakly represented in logs; consequence is underestimation of practical workload and exception handling.

The most challenging facet is *Human*. With the available methods, tacit judgement and informal handoff logic are difficult to assess. A stronger design would combine log analytics with targeted qualitative validation (short observations and focused interviews around critical cases).

6. **Invisible work and system design implications.** The completed invisible-work table is in Appendix B, Table 9, and includes three activities from the workflow diagram that are not represented explicitly in the log. For each activity, the non-logging reasons combine *technical* constraints (events happen outside core systems), *social* factors (verbal/ad hoc coordination), and *economic* constraints (documentation burden under time pressure). A model based only on log traces therefore overestimates formal linearity and underestimates coordination work.

Consequences by RQ are direct: RQ1 may misclassify bottlenecks, RQ2 may underestimate communication-risk pathways, and RQ3 may train on simplified patterns that ignore real coordination load. This directly parallels Part I, where interview and observation produced a structured *say-do gap*. In both parts, *work-as-recorded* is valuable but incomplete.

In Chapter 9 terms, this is a *formalisation continuum* problem: complete formalisation is neither realistic nor always desirable (Guide to Health Informatics 2026b). This aligns with declarative workflow thinking, where robust support balances *flexibility* and *constraints* instead of enforcing one rigid path (Aalst et al. 2009). A concrete design change is lightweight optional contextual events (consultation, reprioritisation reason, communication intent) with low-friction UI defaults, periodic review, and clear limits on secondary use to mitigate documentation burden and surveillance effects.

A Supporting Tables for Part I

This appendix contains supplementary Part I material referenced from Deliverables 1–4.

Table 5: Observation guide used during Deliverable 2

Focus	Observation prompt	Evidence target
Sequence and boundary	Which event marks start/end, and what is the observed order of actions?	Event IDs, timestamps, sequence notes
Tools and artefacts	Which tools/resources are used at each step?	Tool field, context notes
Interactions and coordination	Does any interaction with other people occur?	Actor field, coordination notes
Deviations from interview baseline	Which actions are skipped, added, reordered, or relabeled relative to A01-A09?	Related activity ID + delta notes
Adaptation and recovery	Which workaround or exception handling occurs under time/resource pressure?	Context notes for exception events

Source: Zajac (2026) and Guide to Health Informatics (2026a): interview + observation + process-tracing principles adapted to this routine study.

Note: Field notes were split into descriptive and reflexive notes; only descriptive notes were transcribed into the submitted CSV event log.

B Supporting Tables and Code for Part II

This appendix contains full Part II evidence tables and the custom method code excerpt referenced from Deliverable 6.

Table 6: Selected methods from Part 3

Method	Primary facets	RQ link	Why selected
M2	Data, Source, System	RQ1-3	Baseline completeness check before further interpretation.
M6	Data, Task, Human	RQ1, RQ3	Captures urgency and temporal representativity in one method.
M8 (15 min)	Task, System, Human	RQ2	Directly tests safety-critical communication conformance.
M10	Data, System, Task	RQ1	Combines TAT metrics with batch-logging sensitivity.
M11	Human, Task, System	RQ1, RQ3	Adds workload and actor-distribution context.

Table 7: Key results and interpretation

Method	Key result	Interpretation (incl. expectation check)	Main facet(s)
M2	High missingness in context attributes (e.g., <code>study_uid</code> 134/150).	As expected; process-flow analysis is possible, but context-dependent inference is limited.	Data, Source
M6	Urgency skew: 15 stat/urgent vs 2 routine; shift imbalance (Day 11, Night 6, Evening 0).	As expected; representativity is uneven, so generalisation should be cautious.	Data, Task, Human
M8	16 pass, 1 fail (critical case missing communication event).	Worked as intended; even one failed critical case is operationally important for RQ2.	Task, System, Human
M10	Batch logging suspected in 6/17; mean total TAT 147.93 min raw vs 198.50 min clean.	Most surprising output; RQ1 conclusions are highly sensitive to timestamp credibility.	Data, System, Task
M11	Events per role uneven (radiographer 64; resident 7; system actor high concentration).	As expected; workload concentration affects operational and modelling interpretation.	Human, Task, System

Table 8: Facet mapping for Batch Logging Credibility

Facet	Involvement	Justification
Data	++	Timestamp integrity is the direct object of assessment.
Source	++	Logging practice determines whether entries are real-time or delayed.
System	+	Interface/workflow design can incentivise retrospective entry.
Task	+	Timeliness-sensitive tasks are directly affected by timestamp quality.
Human	+	Workload pressure shapes documentation behavior.

B.1 Custom DQ Assessment Method Code Excerpt

```

1 def my_custom_dq_method(df: pd.DataFrame):
2     rows = []
3     for case_id, cdf in df.groupby('case_id'):
4         n = len(cdf)
5         u = cdf['event_time'].nunique()
6         dup = n - u
7         dur = ((cdf['event_time'].max() - cdf['event_time'].min()).total_seconds() / 60) if n > 1
            else 0
8         score = (2 if dup >= 3 else 1 if dup > 0 else 0) + (2 if n >= 5 and dur < 2 else 0) +
            (1 if (u/n) < 0.5 else 0)
9         risk = 'high' if score >= 4 else ('medium' if score >= 2 else 'low')
10        rows.append({'case_id': case_id, 'risk_level': risk, 'risk_score': score})

```



```

11 out = pd.DataFrame(rows)
12 flagged = out[out['risk_level'].isin(['high', 'medium'])]
13 return {'case_flags': out,
14         'summary_metrics': {'total_cases': len(out), 'flagged_case_rate_pct': round(100*
15         len(flagged)/len(out), 2)},
16         'examples': flagged.head(3).to_dict('records')}

```

Table 9: Invisible work not represented explicitly in the event log

Activity from diagram	Why not logged?	Facet	Consequence	Capture option
Radiologist-to-radiologist consultation	Verbal/ad hoc interaction	Human, Task	Understates collaborative interpretation work	Optional peer-consult event
Clarification with ordering clinician	Often outside core system	System, Source	Communication-delay attribution may be misleading	Linked communication event
Informal reprioritisation/handoff	Local coordination, not formal workflow step	Task, Human	Bottleneck and fairness analyses miss queue management	Priority-adjusted event with reason code

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